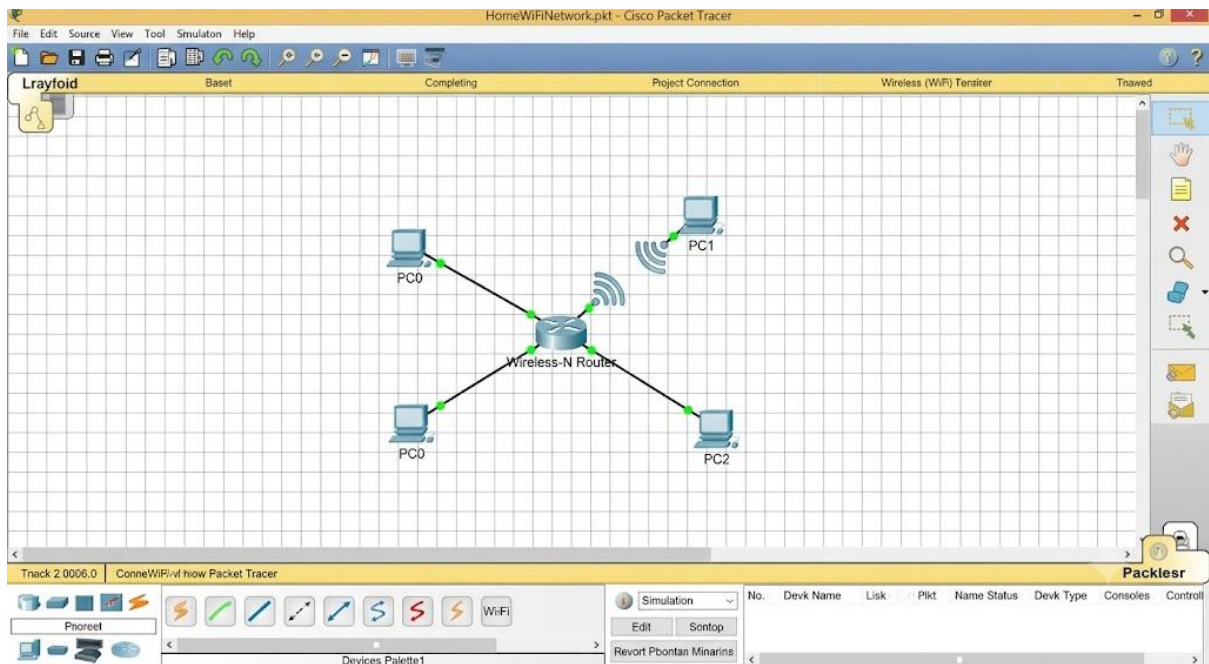


TASK-1



Steps to Create HomeWiFiNetwork.pkt

Step 1: Open Cisco Packet Tracer

- Launch **Cisco Packet Tracer**.
- Select **File** → **New**.

Step 2: Add a Wireless Router

- From the bottom-left menu, click **Network Devices**.
- Select **Wireless Devices**.
- Drag **Wireless Router (WRT300N/Home Router)** into the workspace.

Step 3: Add Three PCs

- Click **End Devices**.
- Drag **PC** three times into the workspace.
- Rename them:
 - **PC-1**
 - **PC-2**
 - **PC-3**

Step 4: Connect the PCs

If using Wi-Fi:

- Open each PC.
- Go to **Physical** tab.
- Turn **Power OFF**.
- Insert the **Wireless Network Adapter (WPC300N/PT-HOST-NM-1W)** if needed.
- Turn **Power ON**.
- Connect each PC to the router's Wi-Fi SSID.

OR

If using Ethernet:

- Click **Connections**.
- Select **Copper Straight-Through Cable**.
- Connect:
 - PC-1 → Router LAN Port 1
 - PC-2 → Router LAN Port 2
 - PC-3 → Router LAN Port 3

Step 5: Configure the Router

- Click the wireless router.
- Ensure:
 - **SSID:** HomeWiFi
 - **DHCP:** Enabled
 - **IP Address:** 192.168.1.1

Step 6: Configure the PCs

On each PC:

- Desktop → **IP Configuration**
- Select **DHCP**

The PCs should receive addresses similar to:

Device IP Address

PC-1 192.168.1.100

PC-2 192.168.1.101

PC-3 192.168.1.102

Gateway:

192.168.1.1

Step 7: Test Connectivity

Open **Desktop → Command Prompt** on PC-1.

ping 192.168.1.101

You should receive:

Reply from 192.168.1.101:

bytes=32 time<1ms TTL=128

TASK-2

Static IP Address Configuration for Cisco Packet Tracer

Network Addressing Plan

Device	IP Address	Subnet Mask	Default Gateway
Wireless Router	192.168.10.1	255.255.255.0	N/A
PC-1	192.168.10.2	255.255.255.0	192.168.10.1
PC-2	192.168.10.3	255.255.255.0	192.168.10.1
PC-3	192.168.10.4	255.255.255.0	192.168.10.1

Step 1: Configure the Wireless Router

1. Click the **Wireless Router**.
2. Go to **GUI** or **Config**.
3. Set:
 - **Router IP Address:** 192.168.10.1
 - **Subnet Mask:** 255.255.255.0
4. (Optional) Disable **DHCP** if you are using only static IP addresses.

Step 2: Configure PC-1

1. Click **PC-1**.
2. Select **Desktop** → **IP Configuration**.
3. Choose **Static**.
4. Enter:

Setting	Value
IP Address	192.168.10.2

Setting	Value
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

Step 3: Configure PC-2

1. Click **PC-2**.
2. Open **Desktop** → **IP Configuration**.
3. Select **Static**.
4. Enter:

Setting	Value
IP Address	192.168.10.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

Step 4: Configure PC-3

1. Click **PC-3**.
2. Open **Desktop** → **IP Configuration**.
3. Select **Static**.
4. Enter:

Setting	Value
IP Address	192.168.10.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

Subnet Mask (All): 255.255.255.0

Default Gateway: 192.168.10.1

Step 5: Test Connectivity

On **PC-1**, open **Desktop → Command Prompt** and run:

```
ping 192.168.10.3
```

Expected output:

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

You can also test:

- ping 192.168.10.4 (PC-3)
- ping 192.168.10.1 (Router)

If the configuration is correct, all devices will communicate successfully because they are on the same **192.168.10.0/24** subnet.

TASK-3

Testing Connectivity in Cisco Packet Tracer

After assigning the static IP addresses, test connectivity between all devices using the **Command Prompt** on each PC.

IP Address Table

Device IP Address

Router **192.168.10.1**

PC-1 **192.168.10.2**

PC-2 **192.168.10.3**

PC-3 **192.168.10.4**

Step 1: Test from PC-1

1. Click **PC-1**.
2. Go to **Desktop → Command Prompt**.
3. Run the following commands:

```
ping 192.168.10.3
```

```
ping 192.168.10.4
```

```
ping 192.168.10.1
```

Expected Output:

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
```

Step 2: Test from PC-2

Open **Desktop → Command Prompt** and run:

```
ping 192.168.10.2
```

```
ping 192.168.10.4
```

```
ping 192.168.10.1
```

Expected Output:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Step 3: Test from PC-3

Open **Desktop** → **Command Prompt** and run:

ping 192.168.10.2

ping 192.168.10.3

ping 192.168.10.1

Expected Output:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Connectivity Test Results

Source Device Destination		Result
PC-1	PC-2 (192.168.10.3)	✓ Success
PC-1	PC-3 (192.168.10.4)	✓ Success
PC-1	Router (192.168.10.1)	✓ Success
PC-2	PC-1 (192.168.10.2)	✓ Success
PC-2	PC-3 (192.168.10.4)	✓ Success
PC-2	Router (192.168.10.1)	✓ Success
PC-3	PC-1 (192.168.10.2)	✓ Success

Source Device Destination		Result
PC-3	PC-2 (192.168.10.3)	✓ Success
PC-3	Router (192.168.10.1)	✓ Success

Conclusion

The **ping** test confirms that all three PCs can successfully communicate with one another and with the wireless router. Since all devices are configured with **static IP addresses on the same subnet (192.168.10.0/24)** and use the correct **default gateway (192.168.10.1)**, the network is functioning correctly.